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Review

# Epithelial-Mesenchymal Plasticity in Cancer Progression and Metastasis

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**Epithelial-to-mesenchymal transition (EMT) and its reversed process, mesenchymal-to-epithelial transition (MET), are fundamental processes in embryonic development and tissue repair but confer malignant properties to carcinoma cells, including invasive behavior, cancer stem cell activity, and greater resistance to chemotherapy and immunotherapy. Understanding the molecular and cellular basis of EMT provides fundamental insights into the etiology of cancer and may, in the long run, lead to new therapeutic strategies. Here, we discuss the regulatory mechanisms and pathological roles of epithelial-mesenchymal plasticity, with a focus on recent insights into the complexity and dynamics of this phenomenon in cancer.**

EMT is a crucial cellular program that enables polarized epithelial cells to transition toward a mesenchymal phenotype with increased cellular motility. EMT plays essential functions during embryonic development by facilitating gastrulation, neural crest delamination, and the generation of diverse cell and tissue types (Nieto et al., 2016). On the other hand, EMT is also aberrantly activated under pathological conditions including organ fibrosis and cancer. To acquire an invasive phenotype for metastatic progression in cancer, carcinoma cells exploit EMT to facilitate its dissociation from primary tumor and dissemination into blood circulation. EMT also endows tumor cells with enhanced stemness and increased resistance to immune clearance and various therapeutic assaults (Nieto et al., 2016) (Figure 1).

Despite its seemingly simple definition, EMT is an extremely complex and diverse process, activated by pleiotropic intrinsic and extrinsic factors and finely regulated temporally and spatially (Lamouille et al., 2014). Depending on the extracellular environment and the state of the tissue, cells can exhibit distinct modes of EMT that subsequently give rise to vastly different phenotype readouts, including behaviors in cancer progression and metastasis. EMT is also highly dynamic and plastic with a wide spectrum of intermediate states in both development and diseases. This high level of plasticity allows the process to be readily reversible to enable the development of specialized organs in embryogenesis and the formation of macrometastasis during malignant progression.

## Cellular and Molecular Hallmarks of Epithelial-to-Mesenchymal Transition

The epithelial-to-mesenchymal transition (EMT) programs under different biological contexts, in either development or disease conditions, share many common hallmarks (Nieto et al., 2016). Together with the loss of apical-basal polarity and the disassembly of the epithelial cell-cell contacts including tight junctions (TJs), adherens junctions (AJs), and desmosomes, the actin cytoskeletal architecture in the cells undergoing EMT is reorganized, allowing them to acquire a spindle-shaped mesenchymal morphology and gain motility by forming protrusions. Furthermore, cells with EMT features are often able to degrade and invade their basal extracellular matrix by expressing matrix metalloproteinases (MMPs). These characteristic cellular hallmarks are accompanied by a series of molecular readouts. The expression or function of epithelial genes such as E-cadherin, certain cytokeratins, and zona occludens 1 (ZO-1) is lost during the transition, while the expression of genes that define the mesenchymal phenotype, such as vimentin, fibronectin, N-cadherin, and β1 and β3 integrins, becomes elevated. In order to orchestrate this complex program, a group of core EMT transcription factors (EMT-TFs) together with epigenetic, post-transcriptional, and post-translational regulators is activated in response to pleiotropic signals (Figure 2, discussed in later sections).

Despite these conserved characteristics, EMT demonstrates great complexity and diversity depending on the cell and tissue context. Instead of a binary decision, developing evidence suggests that EMT represents a spectrum of states with intrinsic plasticity and proceeds in a stepwise manner through different intermediates. Therefore, the cellular and molecular descriptors of EMT should be evaluated in a context-dependent manner. Using both *in vitro* and *in vivo* models, partial EMT states exhibiting features of both epithelial and mesenchymal cells have been noted in many developmental and disease processes (Nieto et al., 2016; Pastushenko et al., 2018; Yu et al., 2013).

## EMT Spectrum in Development and Cancer

During development, EMT plays a crucial role enabling cells of epithelial origin to migrate to long distances to contribute to the formation of different tissues and organs. These cells are believed to be able to migrate individually or collectively in a coordinated manner. In this process, migratory cells harboring different degrees or combinations of epithelial and mesenchymal features display an array of migratory behaviors (Figure 1A). Single-cell migration usually requires a more complete EMT with reduced cell adhesion, loss of apical-basal polarity, gain of front-rear polarity, and increased individual motility (Friedl and Mayor, 2017). In collective migration, multiple cells migrate in the same direction at a similar speed. Although it was previously believed that groups of cells migrate collectively as epithelial cells, more recent evidences suggest that a wide spectrum of cell adhesion strength and EMT states can be found in the

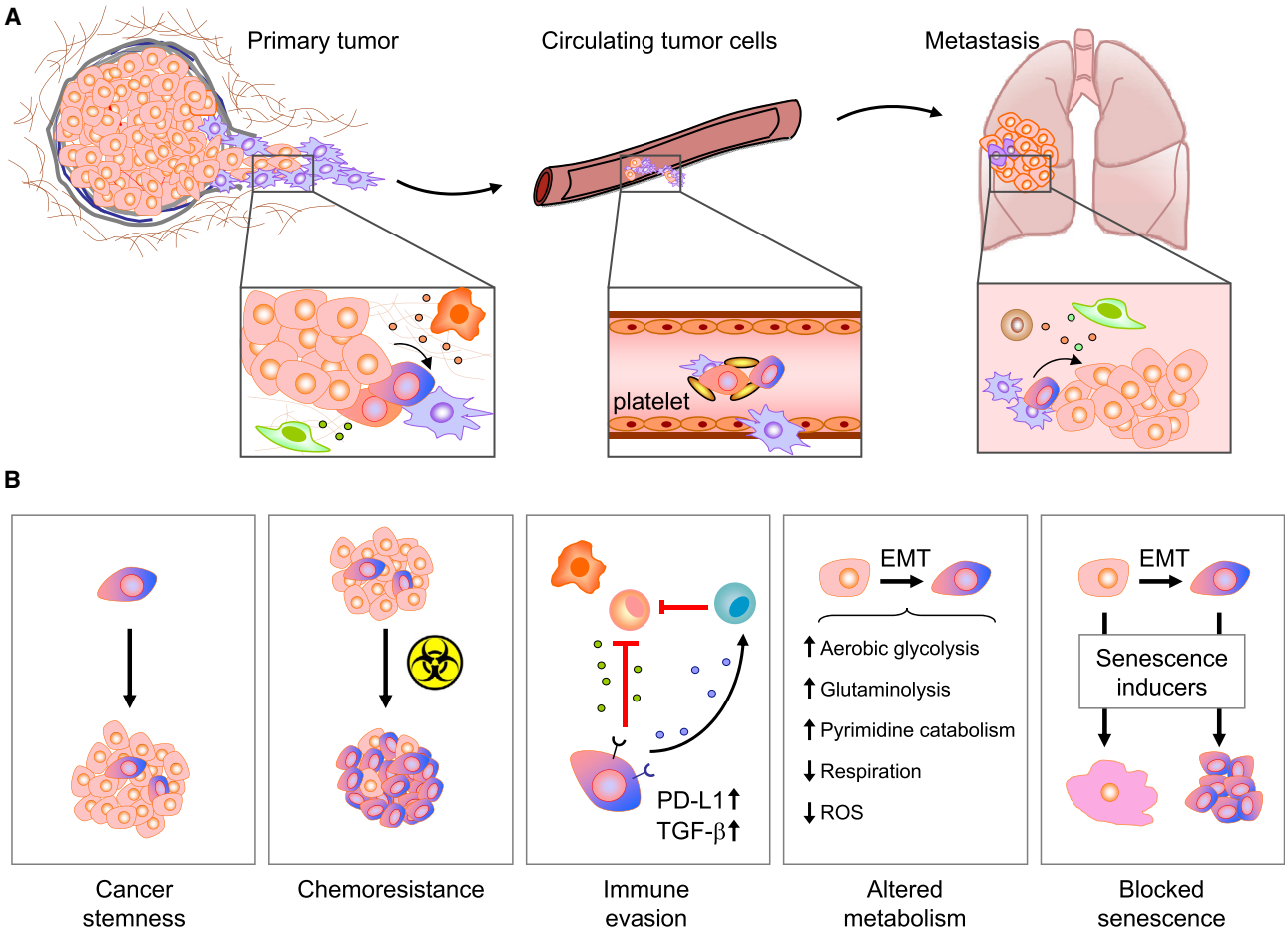


**Figure 1. The Diverse Functions of EMT/MET in Cancer Progression and Metastasis**

(A) In primary tumor, cancer cells at the invasive front undergo EMT under the influence of EMT-inducing growth factors produced by tumor cells or stromal cells. Based on different degrees of transition to the mesenchymal-like state, tumor cells invade through the basement membrane and surrounding tissues using a variety of movement modes, such as collective migration or individual cell invasion. During circulation, EMT can be induced by platelet-produced TGF-β and other signals (Labelle et al., 2011). Upon reaching the distant organs, tumor cells undergo MET under the influence of stromal cells, such as fibroblasts, endothelial cells, and myeloid progenitor cells, and initiate outgrowth to form metastatic lesions.

(B) In addition to promoting metastasis, EMT also induces a variety of malignant features in cancer cells, such as cancer stemness, chemoresistance, immune evasion, altered metabolism (Dong et al., 2013b), and blocked senescence (Ansieau et al., 2008).

migrating clusters (Friedl and Mayor, 2017). Leader cells, localized at the front of the migrating group, undergo partial EMT and gain mesenchymal phenotype with altered polarity and dynamic actin-based protrusive structures to drive migration. At the same time, they retain some epithelial characteristics and remain attached to their neighbors (Mayor and Etienne-Manneville, 2016). The follower cells maintain their apical-basal polarity and intact junctions and migrate through the pulling force generated by leader cells. This phenotype is observed in collective cell migration in embryonic development of various organisms including the development of posterior midgut in *Drosophila*, heart laterality and prechordal plate positioning of zebrafish, mesendoderm migration of *Xenopus*, development of the edge of blastoderm in avian embryo, and gut morphogenesis in mouse (Ocaña et al., 2017; reviewed by Scarpa and Mayor, 2016).

In addition to normal development, EMT is activated aberrantly in cancer and other pathologies such as fibrosis. As in physiological EMTs, pathological EMT also shows great complexity depending on the tissue context. The expression and function of different EMT inducers vary considerably across different cancer types, and the exact function of EMT in cancer progression is still actively debated. For example, using genetic mouse models with conditional gene expression or knockout as well as cell lines with altered gene expression, EMT transcription factors SNAI1 and TWIST1 were found to promote metastasis in breast cancer (Tran et al., 2014; Xu et al., 2017; Yang et al., 2004) but were shown to be dispensable for metastasis in a pancreatic cancer model with knockout of SNAI1 or TWIST1 (Zheng et al., 2015). In contrast, ZEB1 is a strong promoter of metastasis in the same pancreatic cancer model (Krebs et al., 2017) (Figure 3A). These differences may be caused by the different cancer environment and experimental models used, suggesting that EMT inducers may function in a tumor-type-specific manner. Similarly, based on clinical data analysis and experimental tumor models, ZEB2 has been shown to have opposing functions in different cancer types, where it is associated with

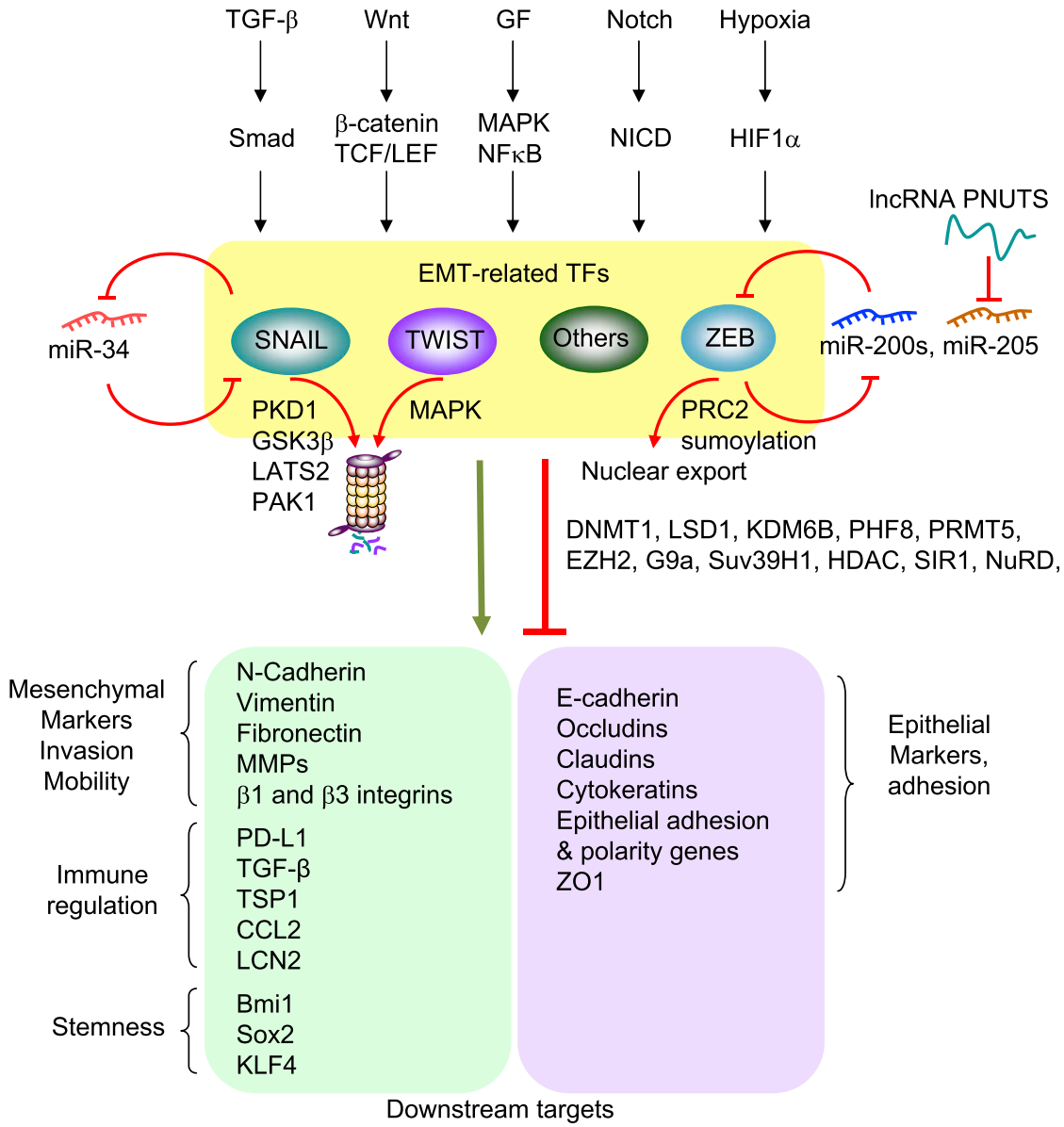


**Figure 2. Multi-Layer Regulatory Network of EMT**

A number of signaling pathways are known to induce EMT-related transcription factors, including the Snail, Twist, and Zeb families and many other transcription factors, which in turn collaborate with a series of epigenetic regulators to induce a transcriptional program that mediates the downstream biological effects of EMT. These EMT-TFs are further regulated at the post-transcriptional and post-translational level by non-coding RNAs and protein stability and/or localization control mechanisms.

metastasis in ovarian, gastric, and pancreatic tumors (Elloul et al., 2005; Imamichi et al., 2007; Rosivatz et al., 2002) but reduces aggressiveness in melanoma by regulating MITF levels to activate melanocyte differentiation (Caramel et al., 2013; Denecker et al., 2014). The diverse combinations of activating signals and co-factors present in different tissues help determine the execution and outcome of the EMT program. Furthermore, cancer cells frequently undergo a partial or transient EMT where various combinations of epithelial and mesenchymal markers co-exist through different intermediate states. Recent studies identified multiple subpopulations of cells associated with different EMT stages from primary skin and mammary tumors that display distinct chromatin landscapes and gene expression signatures (Pastushenko et al., 2018). Similar hybrid states of EMT, with mixed epithelial and mesenchymal characteristics, have also been observed in circulating tumor cells (CTCs) of breast cancer patients (Yu et al., 2013). These findings are also supported by a series of *in vitro* studies confirming the co-expression of epithelial and mesenchymal markers and stepwise transition in breast, ovarian, and lung cancer cell lines (Bierie et al., 2017; Huang et al., 2013; Zhang et al., 2014). Overall, EMT in cancer exhibits great diversity, which may reflect the

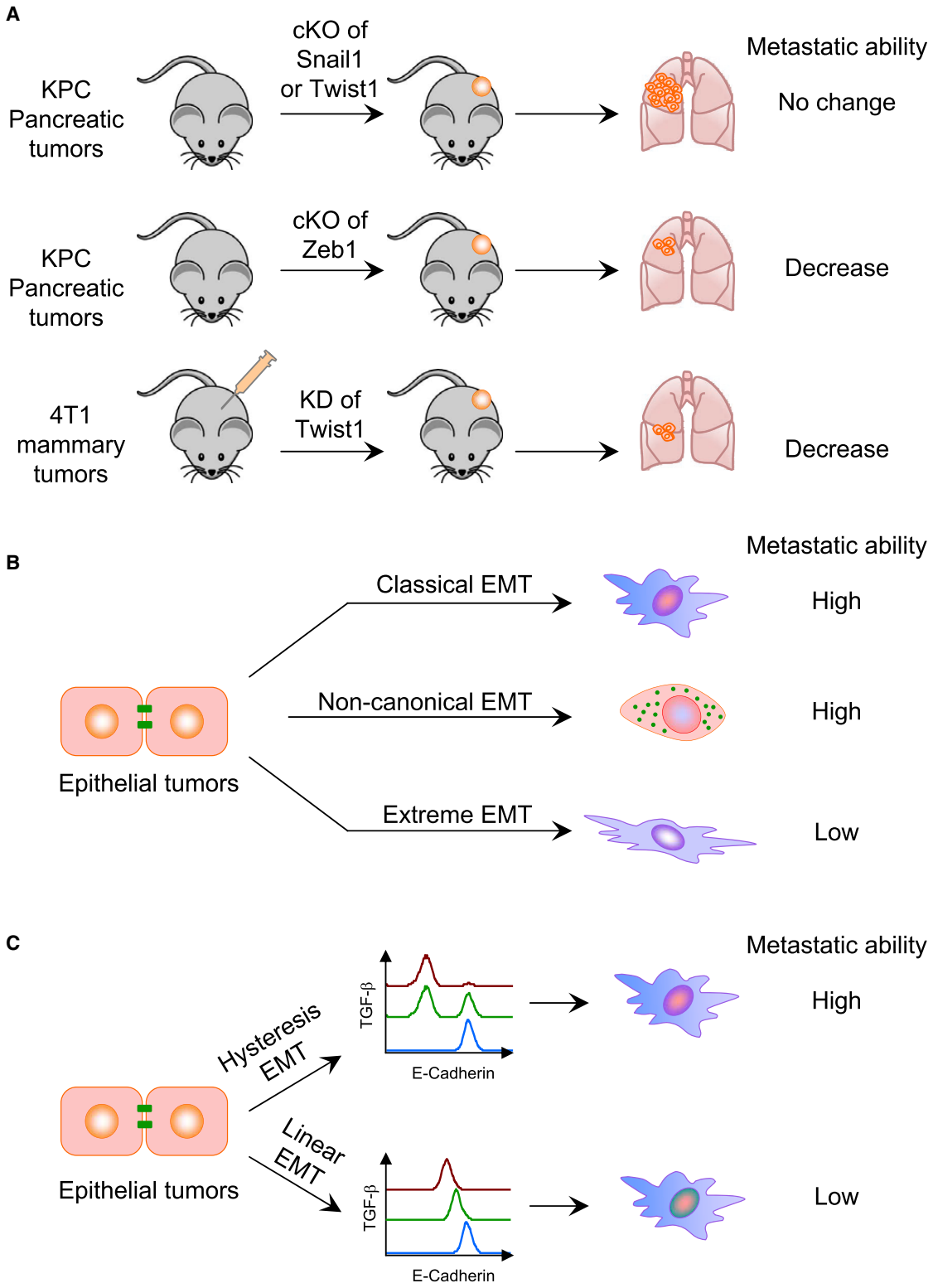


**Figure 3. The Pathological Impact of EMT Is Influenced by Cellular Context and Transitional Mechanisms and Dynamics**

This schematic diagram illustrates some examples of the diversity of EMT and its biological consequences.

(A) Genetic deletion of EMT-TFs Snai1 and Twist1 does not reduce metastasis in the KPC model of mouse pancreatic cancer. In contrast, Zeb1 deletion significantly reduces lung metastasis in the same pancreatic cancer model, and knockdown of Twist1 inhibits metastasis of allograft 4T1 mammary gland tumors.

(B) Classical EMT, which is often driven by EMT-TFs and involves the downregulation of typical epithelial markers and upregulation of mesenchymal markers, promotes cancer metastasis. However, when cancer cell enter an extreme EMT state, the cells may become terminally differentiated or undergo cell death,

*(legend continued on next page)*

fact that EMT can be induced by diverse extracellular signals and finely regulated at different levels. Different hybrid or intermediate EMT status may also have distinct connections with increased tumor stemness, metastatic ability, and resistance to therapy (Nieto et al., 2016).

## Transcriptional Control of EMT

The cellular transdifferentiation from epithelial to mesenchymal states is mediated by key transcription factors that serve as master regulators of cell-cell adhesion, cell polarity, and motility. They repress the genes associated with the epithelial phenotype and induce the expression of mesenchymal genes, ultimately leading to the cellular hallmarks of EMT. Major EMT-inducing transcription factors include zinc-finger-binding transcription factors SNAI1 and SNAI2, the basic helix-loop-helix (bHLH) factors TWIST1 and TWIST2, and the zinc-finger E-box-binding homeobox factors ZEB1 and ZEB2 (Figure 2) (Stemmler et al., 2019). These factors exhibit distinct expression profiles and contributions to EMT depending on cell or tissue type.

SNAI1 and SNAI2 bind to E box sequences in the promoter region of *CDH1* and directly repress its transcription by recruiting the polycomb repressive complex (Batlle et al., 2000; Cano et al., 2000; Herranz et al., 2008). SNAI1 also has a well-established function in repressing the expression of genes regulating TJ formation and apical-basal polarity (Ikenouchi et al., 2003; Whiteman et al., 2008). In addition, SNAI1 can function as a transcription activator to induce mesenchymal genes directly (Hsu et al., 2014; Rembold et al., 2014). Transforming growth factor β (TGF-β), WNT, and Notch pathways, as well as growth factors that act through RTKs, can all activate *SNAI1* expression depending on the physiological context (Peinado et al., 2007). SNAI1 and SNAI2 also cooperate with other transcription regulators to control gene expression (Lamouille et al., 2014; Peinado et al., 2007). Both SNAI1 and SNAI2 play critical roles in the induction of EMT during embryonic development and cancer progression. For example, SNAI1 and SNAI2 enhance the development and migration of the neural crest and facilitate the fusion of palate (Aybar et al., 2003; del Barrio and Nieto, 2002; Martínez-Alvarez et al., 2004; Nieto et al., 1994). In cancer, their expression leads to decreased E-cadherin level, enhanced tumor cell invasion, and metastatic phenotypes in mouse tumor models and cell line studies and is associated with poor prognosis in patients with breast, colorectal, and hepatocellular carcinoma (Blanco et al., 2002; De Craene et al., 2005; Shioiri et al., 2006; Tran et al., 2014; Yook et al., 2006). For example, using multiple genetic breast cancer models with inducible SNAI1 transgene or SNAI1 conditional knockout, it was demonstrated that SNAI1 expression is required for breast tumor metastasis to the lung (Tran et al., 2014).

TWIST1 and TWIST2 belong to the bHLH transcription family that functions as master regulators of a wide array of developmental and pathological processes. TWIST downregulates epithelial genes including E-cadherin and activates expression of mesenchymal genes such as N-cadherin and vimentin (Lamouille et al., 2014). In particular, TWIST-induced suppression of E-cadherin transcription is indirect and mediated by its transcriptional activation of SNAI2, as SNAI2 knockdown blocks the ability of TWIST to activate EMT in mammary cells (Casas et al., 2011). TWIST1 plays essential roles in the EMT process involved in both embryonic morphogenesis and cancer metastasis. It serves as a major regulator of dorsoventral patterning in *Drosophila* and cranial neural tube closure in mouse (Chen and Behringer, 1995; Stathopoulos and Levine, 2002). *Twist* expression can be activated by hypoxia-inducible factor 1α (HIF1α) transcription factor under hypoxic conditions to promote EMT and metastasis (Yang et al., 2008). Using xenograft and transgenic tumor models, it has been shown that TWIST1 is essential for tumor cell dissemination and metastasis in breast cancer and squamous cell carcinoma (SCC), although turning off its expression is required in distant organs to allow subsequent metastases formation (Tsai et al., 2012; Xu et al., 2017; Yang et al., 2004). TWIST overexpression also correlates with cancer invasiveness and metastasis in patients (Lee et al., 2006; Yang et al., 2004).

ZEB1 and ZEB2, two members of the Zeb transcription factor family, directly bind to the E-box elements and repress the expression of E-cadherin (Comijn et al., 2001; Eger et al., 2005). In addition to E-cadherin, ZEB proteins also directly downregulate the expression of TJ genes and components of cell polarity complex to drive EMT (Aigner et al., 2007; Spaderna et al., 2008; Vandewalle et al., 2005). ZEB1 can switch from repressor to activator by interacting with proteins such as YAP1 (Lehmann et al., 2016) or co-activators PCAF and p300 (Postigo, 2003). They also increase the expression of the mesenchymal proteins vimentin and N-cadherin (Bindels et al., 2006; Vandewalle et al., 2005). Expression of ZEB proteins can be induced by TGF-β, Wnt signaling, and other growth factors that activate Ras-MAPK signaling (Lamouille et al., 2014; Peinado et al., 2007). During embryonic morphogenesis, ZEB proteins are essential for neural differentiation and neural crest development (Postigo and Dean, 2000; Stryjewska et al., 2017). In cancer context, ZEB1 and ZEB2 have been experimentally shown to promote cell migration and invasion in breast and colorectal cancer cells (Comijn et al., 2001; Spaderna et al., 2008; Vandewalle et al., 2005). A recent study also provides direct genetic evidence that ZEB1 expression is indispensable for efficient invasion and metastasis in a mouse pancreatic cancer model (Krebs et al., 2017). Interestingly, ZEB1 and ZEB2 show opposing functions in melanoma development, where ZEB2 acts as a tumor suppressor by activating MITF-dependent melanocyte differentiation, while ZEB1 drives melanoma initiation and progression. A switch in the expression from ZEB2 to ZEB1 occurs during malignant progression of melanoma (Caramel et al., 2013; Denecker et al., 2014). Additionally, opposite effects of ZEB1 and ZEB2 have been noted in the regulation of the TGF-β/BMP signaling pathway in different cell lines (Postigo, 2003). These observations highlight the critical importance of tissue context to the function of EMT-TF.

leading to reduced metastasis. In some other instances, EMT is driven by non-canonical pathways, such as internalization of E-cadherin and other post-translational alteration of EMT-related effectors, but still leads to increased metastatic ability in cancer cells.
(C) EMT can occur through hysteresis or liner (non-hysteresis) dynamics, as reflected by bimodal or gradual reduction of E-cadherin expression. Such different dynamics may result in different metastatic ability of affected cancer cells, despite similar appearance of the mesenchymal state at the end point of the transition.

In addition to these well-established core regulators, EMT is also coordinated by a diverse group of other transcription factors, including several Fox transcription factors, PRRX1, HGMA2, and Sox transcription factors, that can induce similar changes in certain tissue contexts (Lamouille et al., 2014; Nieto et al., 2016). These EMT-associated factors may play a facilitating role to promote EMT in more specific contexts. The exact regulation and roles in different EMT contexts of these more recently identified EMT-associated factors are less well defined and require more investigation.

## Regulation of EMT beyond Transcription Factors

To perform their elaborate tissue-specific or stage-specific functions, EMT-TFs are intricately regulated at various levels encompassing epigenetic modifications, alternative splicing, modulation by miRNAs and other non-coding RNAs, translational control, and post-translational modifications. These regulatory components form crosstalk and collaborate with each other to integrate cues from the microenvironment to coordinate the phenotypic reprogramming of epithelial cells (Figure 2).

### Non-Coding RNA-Mediated Regulation

MicroRNAs (miRNAs) can directly regulate the expression of EMT transcription factors. The miR-200 family of five miRNAs and miR-205 repress the expression of Zeb1/2 (Bracken et al., 2008; Burk et al., 2008; Gregory et al., 2008; Korpal et al., 2008; Park et al., 2008). miR-200s and Zeb1/2 form a double-negative feedback loop where ZEB1/2 inhibits the expression of miR-200 miRNAs and miR-200 suppresses ZEB1 expression (Bracken et al., 2008; Park et al., 2008) (Figure 2). During EMT, miR-200 and miR-205 miRNAs are dramatically downregulated, leading to the expression of ZEB transcription factors. Similarly, miR-1 and miR-200 can inhibit SNAI2 expression, and SNAI2 directly represses the transcription of miR-1 and miR-200 (Liu et al., 2013). Such double-negative feedback mechanism allows robust enforcement of epithelial and mesenchymal states under different microenvironment conditions, such as high or low level of TGF-β. The expression of SNAI1 can be similarly repressed by miR-34 (Siemens et al., 2011), miR-203 (Moes et al., 2012), and miR-29b (Ru et al., 2012) in colorectal cancer, breast cancer, and prostate cancer cells, respectively. The tightly interconnected negative feedback loops formed by pairs of miRNAs and EMT-TFs efficiently reinforce the system and ensure rapid and robust response to minimal signals (Celià-Terrassa et al., 2018). The redundancy of miRNA control observed in the targeting of different EMT-TFs may further allow flexible regulation under different physiological and pathological conditions. Besides regulating master transcription factors, miRNAs can also modulate EMT by directly acting on epithelial or mesenchymal genes (Lamouille et al., 2014). For example, miR-9, which is upregulated in breast cancer cells, can downregulate E-cadherin expression, leading to increased cell motility and invasiveness (Ma et al., 2010).

Long non-coding RNAs (lncRNAs) also play important roles in regulating EMT. For example, the lncRNA-PNUTS acts as a competitive sponge for miR-205 and promotes EMT by enhancing ZEB expression of breast tumor cells (Grelet et al., 2017). Depletion of lncRNA-HIT, a conserved lncRNA that locates in the Hoxa gene cluster, inhibits TGF-β-induced EMT and cell migration in mammary epithelial and breast cancer cells (Richards et al., 2015). The repressive activity of SNAI1 has also been shown to be mediated by lncRNA HOTAIR, which recruits EZH2, a key component of PRC2 complex and the main writer of chromatin repressive marks, to specific genomic sites (Battistelli et al., 2017). Taken together, the regulated activities of miRNAs and lncRNAs represent a powerful layer of regulation controlling EMT.

### Epigenetic Modifications

Significant epigenetic changes are associated with EMT and are frequently required to mediate the functions of EMT-TFs. Reprogramming of the genome-scale epigenetic landscape occurs during normal development and differentiation and contributes to pathological EMT (Hawkins et al., 2010; McDonald et al., 2011). Interestingly, EMT program is impacted by the epigenetics of the cell of origin. The epigenetic landscape of SCCs derived from hair-follicle stem cells primes these cells to undergo EMT and metastasis, while the interfollicular-epidermis-derived SCCs remain mostly epithelial and non-metastatic, highlighting the importance of chromatin states and cell of origin in dictating tumor progression (Latil et al., 2017).

Over the past decades, specific modifications, including both DNA methylation and histone modifications, and multiple epigenetic regulators have been revealed as key modulators of the EMT program. *CDH1* promoter methylation has been recognized as an important contributor to EMT and a common event in multiple human cancers (Tamura et al., 2000). Supporting these observations, ZEB1 has also been recently reported to recruit DNMT1, the DNA methyltransferase responsible for methylating CpG residues, to *CDH1* promoter to maintain its methylation status and inhibit its transcription (Fukagawa et al., 2015).

Epigenetic modifications also involve histone modifications, which affect gene expression either directly by changing histone-DNA interaction or indirectly via the differential recruitment of chromatin-remodeling complexes or RNA polymerase complex. In TGF-β-induced EMT of non-transformed hepatocytes, the heterochromatic H3K9me2 mark is reduced, whereas the H3K4me3 euchromatic and H3K36me3 transcription elongation marks are increased. These changes seem to be crucial for EMT-driven cell migration and chemoresistance and depend largely on lysine-specific demethylase-1 (LSD1), a subunit of the NuRD complex (McDonald et al., 2011). LSD1 has been shown to interact with SNAI1, suppress *CDH1* expression, and enhance tumor invasion in several studies (Lin et al., 2010). Its activity is further regulated by MOF acetyltransferase, which acetylates LSD1 in epithelial cells and blocks EMT (Luo et al., 2016). Besides the histone demethylase LSD1, other demethylases such as KDM6B (Pereira et al., 2011; Ramadoss et al., 2012) and PHF8 (Shao et al., 2017), as well as methyltransferases PRMT5 (Hou et al., 2008), EZH2/SUZ12 (Battistelli et al., 2017; Herranz et al., 2008), SUV39H1 (Dong et al., 2013a), and G9a (Dong et al., 2012; Si et al., 2015), have all been reported to modulate EMT by either regulating the expression of EMT-TFs (mainly *SNAI1* and *ZEB*) or interacting with these factors to impact downstream gene expression. In addition to methylation, histone acetylation and deacetylation also play important roles in the modulation of EMT. During the transition from epithelial to mesenchymal states of trophoblast stem cells, one of the first

developmental EMT events, histone deacetylase HDAC6 directly deacetylates the promoters of TJ genes, resulting in diminished cell-cell adhesion (Mobley et al., 2017), while histone acetyltransferase CBP acetylates H2A and H2B to maintain the epithelial phenotype of mouse trophoblast stem cells (Abell et al., 2011). SNAI1 induces repressive histone modifications at the *CDH1* promoter through recruiting transcription repressors such as HDACs and SIN3A (Peinado et al., 2004). Similarly, ZEB1 cooperates with the deacetylase sirtuin 1 (SIRT1) at the *CDH1* promoter, leading to the deacetylation of histone H3 and reduced binding of RNA polymerase II, which ultimately leads to enhanced prostate cancer metastasis in mice (Byles et al., 2012). Transcription factors of ZEB and TWIST families also bind and recruit the nucleosome-remodeling deacetylase NuRD complex to their target promoters (Fu et al., 2011). Interestingly, ZEB2/NuRD(MTA1) forms a negative feedback loop with GATA3/NuRD(MTA3). During breast cancer progression, GATA3 and MTA3 are downregulated, leading to elevated ZEB2/NuRD(MTA1) that contributes to metastasis formation (Si et al., 2015). Additionally, MTA3 has also been shown to directly repress *SNAI1* transcription in breast cancer cells in a histone deacetylase-dependent manner (Fujita et al., 2003). Overall, epigenetic modifications are fundamental in determining the expression of key proteins in EMT. Importantly, these modifications are usually reversible and can have critical functions in defining the plasticity of EMT.

## Translational and Post-Translational Control

EMT is also significantly impacted by translational and post-translational control. Enforced expression of Y-box-binding protein-1 (YB-1) induces EMT and promotes metastasis by directly activating cap-independent translation of SNAI1 mRNA as well as other mesenchymal factors in Ras-transformed mammary epithelial cells (Evdokimova et al., 2009).

Besides direct translational regulation, subcellular localization of important EMT players also significantly impacts cell behaviors. For example, EMT can occur by internalization of E-cadherin protein into endosomes, in addition to the more canonical way of transcriptionally repressing epithelial genes (Figure 3B). These different EMT programs can lead to different modes of cell invasion during pancreatic cancer progression as shown in a lineage-labeled mouse model (Aiello et al., 2018).

As an important type of post-translational modification, phosphorylation is essential in determining the stability and localization of key transcription factors including SNAI1, SNAI2, and TWIST. GSK-3β, a key serine/threonine kinase in the canonical WNT signaling pathway, phosphorylates and regulates the activity of SNAI1 in controlling EMT in normal and malignant mammary cells (Bachelder et al., 2005; Yook et al., 2006; Zhou et al., 2004). GSK-3β binds to and phosphorylates SNAI1 at two consensus motifs for ubiquitin-mediated degradation and nuclear export (Zhou et al., 2004). Indeed, several signaling pathways including Wnt, Notch, and nuclear factor κB (NF-κB) signaling increase SNAI1 activity and induce EMT by inactivating GSK-3β-mediated phosphorylation (Sahlgren et al., 2008; Wu et al., 2009; Yook et al., 2006). Additionally, subcellular localization of SNAI1 is also controlled through phosphorylation by several other kinases including LATS2, PAK1, and PKD1 in cancer cells of colon, breast, and prostate origins (Du et al., 2010; Yang et al., 2005; Zhang et al., 2012). Importantly, the phosphorylation and subcellular distribution of SNAI1 are affected by the extracellular environment, with increased phosphorylated and cytoplasmic localization of SNAI1 in suspended cells compared to adhered cells (Domínguez et al., 2003). Pathways and proteins responsible for the nuclear import and export of SNAI1 have also been elucidated (Mingot et al., 2009; Mingot et al., 2013). Multiple ubiquitin ligases have been identified to mediate the degradation of SNAI1 and SNAI2 (Viñas-Castells et al., 2010; Wang et al., 2009; Zheng et al., 2014; Zhou et al., 2004). In addition to SNAI1, the stability of TWIST is also regulated by phosphorylation, which protects it from ubiquitin-mediated degradation when phosphorylated by MAPKs (Hong et al., 2011) (Figure 2). Besides phosphorylation, other post-translational modifications have also been shown to regulate the activity of key EMT-TFs and modulate the EMT decision. Acetylation of SNAI1 protein by CBP prevents the formation of the repressor complex and turns SNAI1 from gene repressor into an activator (Hsu et al., 2014). SNAI1 is also subject to O-GlcNAcylation at Ser 112 under hyperglycemic conditions, which leads to its stabilization (Park et al., 2010). Sumoylation of ZEB2 by PRC2 promotes its export from nucleus and attenuates its transcription factor activity (Long et al., 2005).

## Pleiotropic Roles of EMT in Cancer

### *EMT and MET in Metastasis*

Since its early description in *in vitro* cancer models decades ago, the role of EMT in cancer progression and metastasis *in vivo* has been widely reported but has also remained a topic of vigorous debates. As discussed earlier, tumor cells often undergo an incomplete EMT during metastasis, and recent studies have confirmed the existence of these partial EMT states during tumor progression *in vivo* (Pastushenko et al., 2018; Yu et al., 2013). By overexpressing or downregulating key EMT-TFs, many studies have reported the critical importance of EMT in promoting metastatic phenotypes. For instance, overexpression of SNAI2, together with SOX9 enhances tumorigenic and metastasis-seeding abilities of human breast cancer cells (Guo et al., 2012). ZEB1 can promote colorectal and breast cancer metastasis to liver and lung in mouse xenograft models by suppressing the expression of cell polarity factors (Spaderna et al., 2008). Suppression of TWIST expression in highly metastatic mammary carcinoma cells inhibits their ability to metastasize to the lung (Yang et al., 2004). However, most of these studies relied on cultured cell lines or xenograft models and artificially kept cells in fixed mesenchymal-like states by enforced overexpression of EMT-TFs, which does not accurately reflect the natural EMT process where cancer cells adopt a dynamic spectrum of transitional EMT states *in vivo*. In particular, it is important to use cell-fate mapping to determine whether metastatic lesions are indeed established by cells that have undergone EMT, rather than just correlation between altered levels of EMT-TF expression with metastatic outcomes.

The debate about the functional importance of EMT in metastasis was heightened when two recent studies using genetic mouse models for breast and pancreatic cancer suggested that EMT is dispensable for metastasis (Fischer et al., 2015; Zheng et al., 2015). However, closer examination of the studies revealed cautions that need to be taken into account when

interpreting the results (Aiello et al., 2017; Ye et al., 2017). In particular, the absence of a particular EMT marker such as Fibroblast-Specific Protein 1 (FSP1) or vimentin does not necessarily mean an absence of EMT or partial EMT program. Likewise, because of the functional redundancy and compensation of multiple EMT-TFs, lack of metastatic phenotype after genetic deletion of one EMT-TF does not necessarily nullify the importance of EMT for a particular cancer type. Interestingly, using the same KPC pancreatic cancer model in which SNAI1 and TWIST deletion failed to significantly reduce metastasis (Zheng et al., 2015), genetic depletion of ZEB1 strongly suppresses invasion and metastasis (Krebs et al., 2017), suggesting that EMT-TFs may exhibit variability and tissue specificity in promoting metastasis (Figure 3A). In addition, several other recent studies also provided direct evidence for EMT function under physiological conditions (Beerling et al., 2016; Pastushenko et al., 2018; Rhim et al., 2012; Tran et al., 2014; Tsai et al., 2012; Xu et al., 2017; Ye et al., 2015). Using genetically engineered reporter mouse model to follow the endogenous expression of SNAI1 during cancer progression, it was shown that SNAI1 activation and EMT indeed occur in primary tumor cells that ultimately disseminate and that SNAI1 is essential for the metastasis of mouse mammary carcinoma cells (Tran et al., 2014; Ye et al., 2015). Furthermore, intravital imaging in mice revealed that migratory breast tumor cells undergo EMT to disseminate and reverse to epithelial state upon metastatic outgrowth (Beerling et al., 2016). Multiple tumor subpopulations of different EMT stages with different metastatic potential were also identified from primary mammary and skin tumors by screening a large panel of cell surface markers (Pastushenko et al., 2018). This study also suggested that cells with hybrid epithelial and mesenchymal phenotypes are more efficient in reaching the circulation and forming metastases. Additionally, it is also important to note that EMT exhibits distinct types of dynamics of transition and that different modes of EMT can lead to different metastatic outcomes. For instance, only cells undergoing EMT in a non-linear hysteretic mode, but not non-hysteresis EMT, show increased metastasis in a breast cancer model (Celià-Terrassa et al., 2018) (Figure 3C). Such great diversity of EMT underscores the importance of using caution while interpreting experimental findings when studying the functional contribution of EMT to metastasis.

Although EMT may be crucial for dissemination from primary tumors, the process alone is not sufficient and may even be detrimental for metastatic colonization at distant organs, as tumor cells need to recover some epithelial characteristics to support the outgrowth of metastatic lesions. The clinical observation that most metastases maintain epithelial features led to the proposition of the transient EMT model, which suggests that EMT is involved in invasion, dissemination, and extravasation, but a subsequent MET step is required for metastatic colonization. This model has been validated in several experimental models (Beerling et al., 2016; Chaffer et al., 2006; Korpal et al., 2011; Ocaña et al., 2012; Ruscetti et al., 2015; Tsai et al., 2012). For example, although TWIST1 is sufficient to promote dissemination of tumor cells into circulation in a spontaneous SCC mouse model, its subsequent downregulation is required for metastasis formation (Tsai et al., 2012). Recent studies have shown that MET can be induced by either cell-intrinsic or stromal components (del Pozo Martin et al., 2015; Pattabiraman et al., 2016). For example, an increase in the intracellular levels of cAMP and the subsequent activation of PKA induce MET in mesenchymal human mammary epithelial cells (Pattabiraman et al., 2016). Tumor-associated fibroblasts, endothelial cells, and myeloid progenitor cells can also induce MET (del Pozo Martin et al., 2015; Esposito et al., 2019; Gao et al., 2012) (Figure 1).

### *Cancer Stemness*

The cancer stem cell (CSC) model was postulated several decades ago, proposing that CSCs, a small fraction of malignant cells within a tumor, possess tumor-initiating potential and can differentiate into the great diversity of cell types found within the tumor mass. Accumulating evidence supports the existence of CSC in many prevalent tumor types, yet it is also widely acknowledged that not every cancer follows the CSC model and hierarchy. Moreover, CSC hierarchy is usually not rigid but highly plastic, complicating the identification and isolation of CSCs *in vivo*. Recent studies have linked EMT activation with the acquisition of stem-cell properties and identified EMT as a critical regulator of the CSC (Figure 1B). Experimental overexpression of SNAI1, SNAI2, or TWIST in immortalized or transformed human mammary epithelial cells leads to a stem-cell phenotype with increased sphere-forming ability and expression of stem-cell markers, whereas their inhibition blocks mammary gland reconstitution and tumor-initiating activity (Battula et al., 2010; Guo et al., 2012; Mani et al., 2008; Morel et al., 2008; Vesuna et al., 2009). ZEB1 is also shown to confer stem-cell properties in mouse and human breast, pancreatic, and colorectal cancer cells by repressing stemness-inhibiting miRNAs such as miR-200s, miR-183, and miR-203, which further control the expression of other stem-cell-associated factors including BMI1, SOX2, and KLF4 (Shimono et al., 2009; Wellner et al., 2009). A high expression of EMT signature associates closely with the claudin-low and metaplastic breast cancer subtypes that have stemness features (Taube et al., 2010). This association is also confirmed experimentally in circulating pancreatic tumor cells and metastatic breast cancer cells *in vivo*, which have undergone EMT, express CSC-associated markers, and exhibit stem-cell properties (Lawson et al., 2015; Rhim et al., 2012).

However, the association between EMT and stemness has been challenged by conflicting evidence uncoupling EMT and stemness, suggesting that an extreme EMT may lock cells in fully differentiated states and diminish their plasticity and stem-cell properties (Celià-Terrassa et al., 2012; Ocaña et al., 2012; Schmidt et al., 2015). These studies argue, instead, that the repression of EMT process is required for stemness. To reconcile these seemingly contradictory findings, a unifying model has been put forward that cells in intermediate or hybrid EMT, instead of extreme epithelial or mesenchymal states, are more likely to acquire stemness. According to this hypothesis, the highest degree of plasticity and the highest potential to become stem cells are manifested in cells of intermediate EMT states that possess both epithelial and mesenchymal traits (Jolly et al., 2015; Nieto et al., 2016). This model is supported by several lines of evidence obtained from sphere-formation and tumorigenicity studies using prostate and breast cancer models (Mani et al., 2008; Ocaña et al., 2012; Ruscetti et al., 2015). For instance, research shows that ovarian cancer cells in hybrid phenotypes are multipotent, express markers of other lineages and drive

tumor growth *in vivo* by giving rise to different subsets of hybrid and differentiated epithelial cells (Strauss et al., 2011). Furthermore, different EMT subpopulations isolated from skin and mammary primary tumors demonstrate different clonogenic and differentiation potentials as well as different degree of plasticity (Pastushenko et al., 2018). This model also explains the possible association between hybrid EMT and increased metastasis because a partial transition endows cells with a greater potential for both dissemination and colonization. Although detailed molecular mechanisms connecting EMT with stemness still remains elusive, the link between EMT and stemness suggests that eliminating these cells with high metastatic potential may represent a promising avenue of cancer therapy to target CSCs in patients.

### *Chemoresistance and Immune Suppression*

Emerging evidence suggests that EMT contributes to the resistance to chemotherapy in multiple cancer types and may serve as a potential target for overcoming chemoresistance (Figure 1B). Earlier studies established the connection between EMT and drug resistance by assessing the drug sensitivity of cancer cell lines with altered expression of EMT-TFs (Arumugam et al., 2009; Kurrey et al., 2009; Vega et al., 2004). More recently, two separate studies using genetically engineered mouse models demonstrate that primary and metastatic tumor cells become more resistant to chemotherapeutic drugs in an EMT-dependent manner in both breast and pancreatic cancers (Fischer et al., 2015; Zheng et al., 2015). These studies provide convincing evidence linking EMT to chemoresistance and highlight the potential of targeting EMT in cancer therapy.

EMT has also been shown to contribute to immune suppression and resistance to immunotherapy (Figure 1B). Although various immune cells, such as $CD8^+$ T cells and natural killer (NK) cells, have strong antitumor functions, tumor cells often develop a mechanism to escape immune surveillance and generate an immunosuppressive tumor microenvironment. Immunotherapy based on the blockade of immune inhibitory checkpoints has gained tremendous therapeutic successes against multiple tumor types. However, many cancer types remain refractory to immunotherapy, and development of treatment resistance is often observed in clinics. Activation of EMT may be a key process that affects the function of immune cells in the tumor microenvironment and contributes to immunosuppression and immunoresistance. Several studies have reported that tumor cells undergoing EMT exhibit greater resistance to antitumor immune response. In melanoma, SNAI1-induced EMT accelerates cancer metastasis partly through the induction of regulatory-T-cell-mediated immunosuppression (Kudo-Saito et al., 2009). In breast cancer, acquisition of EMT phenotype in MCF-7 human breast cancer cells was associated with a blockade of cytotoxic T lymphocytes (CTL) activity (Akalay et al., 2013). Furthermore, mammary tumors arising from more mesenchymal carcinoma cell lines exhibiting EMT markers express high levels of PD-L1 and contain within their stroma regulatory T cells, M2 macrophages, and exhausted $CD8^+$ T cells, whereas the epithelial tumors are infiltrated with cytotoxic $CD8^+$ T cells and M1 macrophages. Consequently, epithelial tumors are more susceptible to elimination by immunotherapy compared with the mesenchymal ones (Dongre et al., 2017).

Despite the clear link between EMT and immunosuppression, the precise mechanisms that confer the antitumor resistance still require further investigation. The immune components can be modulated by the cytokines secreted from tumor cells, the production of which may be directly regulated by EMT-TFs (Hsu et al., 2014; Kudo-Saito et al., 2013). Additionally, EMT can directly induce the expression of PD-L1 in carcinoma cells (Chen et al., 2014; Noman et al., 2017). This is also consistent with the findings that EMT signature correlates with tumor types that respond best to anti-CTL-A4 and PD1/PD-L1 treatments and with tumors that have increased expression of multiple immune checkpoint markers (Lou et al., 2016; Tan et al., 2014).

## Conclusions and Perspectives

EMT is essential for normal embryogenesis but has detrimental consequences in promoting cancer progression and metastasis. Despite intense interest and explosive growth in the area of EMT research, we are still far from reaching a complete understanding of the biological mechanism and pathological contribution of epithelial-mesenchymal plasticity in cancer. A wide variety of signals have been reported to activate EMT in various normal and cancerous tissues. A complex network of transcriptional regulators coupled to post-transcriptional and post-translational modifications control the execution of EMT. Despite the wealth of knowledge that has accumulated over the past few decades, there are many mysteries that remain to be resolved. Why do different EMT-TFs and EMT dynamics have distinct contributions to cancer progression and metastasis? What molecular features of intermediate or partial EMT endow the most robust pro-metastatic and stemness properties to cancer cells? How do cancer cells with different EMT status cooperate during different stages of tumor initiation, progression, metastasis, and treatment resistance? Recent developments in sophisticated lineage tracing animal models and new technologies such as high-resolution intravital imaging and single-cell sequencing techniques will greatly facilitate research in these directions.

The pleiotropic functions of EMT associated with cell stemness, metastasis, and therapy resistance highlight great therapeutic opportunities for targeting EMT to enhance the efficacy of chemotherapy, targeted therapy, or immunotherapy. Although tumor invasion and dissemination may be an early event in cancer progression, targeting stemness properties, chemoresistance, and immune suppression conferred by the EMT program could potentially enhance therapy efficacy and reduce recurrence. However, great care must be taken when designing EMT-inhibiting strategies. Because of the plasticity of the program and the need for tumor cells to revert to a more epithelial state to colonize distant organs, inhibition of EMT could inadvertently promote the formation of secondary tumors. In contrast, targeting key control mechanisms of cellular plasticity, such as epigenetic regulators of EMT/MET, may prevent the flexible interconversion of cancer cells between the epithelial and mesenchymal states required for metastasis and sensitize them to stress during cancer progression and therapeutic assaults. Alternatively, instead of blocking EMT or MET, therapeutic agents that push cancer cells toward the extreme EMT states may lead to terminal differentiation or apoptosis of cancer cells,

as has recently been demonstrated in breast cancer and pancreatic cancer (David et al., 2016; Ishay-Ronen et al., 2019).

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